

Universal File Type Converter Web Application

IIT Indore SoC 26

Software Development PS3 Project Proposal

Team Details

Abstract

The management and conversion of diverse file formats present a challenge for users requiring quick transitions between document structures and asset encodings. We propose a universal file type converter web application designed to handle multi-format file transformations securely. The platform leverages a modular architecture to ingest user assets, process formats safely via backend tools, and ensure absolute user data privacy by automatically removing processed files from storage mounts within a fixed timeline.

Team Details

Team Leader

- **Name:** Vidadala Kalyani
- **Roll Number:** 250001077
- **Links:** GitHub — LinkedIn — Discord ID
- **Project Role:** Backend Developer
- **Project Work:** Building the server routing infrastructure, configuring file upload managers, and connecting system conversion utilities to execute the core format transitions.

Other Member

- **Name:** Vithanala Anushka
- **Roll Number:** 250001080
- **Links:** GitHub — LinkedIn — Discord ID
- **Project Role:** Frontend Developer
- **Project Work:** Designing the file upload dashboard layout, constructing the drag-and-drop workspace area, and managing visual progress bars to track file states.

Other Member

- **Name:** Kondamuri Tathwik
- **Roll Number:** 250021010
- **Links:** GitHub — LinkedIn — Discord ID
- **Project Role:** DevOps & Security Engineer
- **Project Work:** Setting up the local Docker container workspace, programming automated file cleanup tasks to enforce user privacy, and managing directory security parameters.

Proposal Details

- **Domain:** Software Development
- **Problem Statement Name:** Universal File Type Converter Web Application
- **Problem Statement Number:** PS 3
- **Preference:** First

Solution Architecture

File Ingestion Pipeline

- Build a responsive user interface containing interactive file picking fields.
- Use backend file handling managers to parse multi-part form payloads safely.
- Implement validation barriers to verify incoming sizes and look for bad extensions.

Core Conversion Processing

- Read target conversion codes selected by the user from the interface modules.
- Trigger isolated runtime background operations on the server to execute translation software.
- Map formatted assets to dedicated download links so users can retrieve their outputs.

Privacy and Security Controls

- Limit server system access strictly to specific processing storage folders.
- Use automated local system task timers to scan storage paths every single minute.
- Enforce absolute data erasure rules by purging every file asset exactly 30 minutes after creation.

Project Timeline

Phase	Dates	Tasks
Setup & Init	June 3 — June 12	• Initialize code repository access frameworks. • Create localized Docker development containers.
Frontend & Routes	June 13 — June 18	• Build drag-and-drop file dashboard interfaces. • Map multi-part target server path routes.
Core Processing	June 19 — June 30	• Code inner child conversion process commands. • Prepare system mid-evaluation working prototype.
Mid-Evaluation	June 30	• Submit operational baseline demo and code modules.
Security Engine	July 1 — July 14	• Code automated task scheduling cleanup routines. • Run format input exception protection routines.
Testing & Deploy	July 15 — August 2	• Run multi-browser responsive check runs. • Host final application platform on cloud services.
Final Submission	August 2	• Project repository, dashboard link, and report due.

Table 1: Development Phases and Milestones

Tech Stack

- **Frontend Layer:** React.js framework, Tailwind CSS styles.
- **Backend Architecture:** Node.js runtime environment, Express.js server frameworks.
- **Platform Utilities:** Multer data parser, ImageMagick, Pandoc, Node-cron scheduler, Docker virtual containers.